

VISHANTH K

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PROFESSIONAL SUMMARY

Aspiring **AI/ML Engineer** with hands-on experience in Machine Learning, Deep Learning, NLP, LLMs, RAG, Computer Vision, Edge AI, and full-stack development. Experienced in building predictive models and intelligent solutions for real-world applications.

EDUCATION

Coimbatore Institute of Technology (CIT)

Integrated M.Sc. in Artificial Intelligence & Machine Learning
CGPA: 7.88/10

Coimbatore, Tamil Nadu

2024 – 2028

Expected Graduation: 2028

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, JavaScript

AI/ML: Machine Learning, Deep Learning, NLP, Computer Vision, LLMs

Data: Pandas, NumPy, Scikit-learn, Matplotlib, Feature Engineering

GenAI: RAG, Prompt Engineering, Vector Search, ChromaDB

Frameworks: PyTorch, FastAPI, Flask, React, Node.js

Edge AI: ONNX, Raspberry Pi, Model Optimization, Sensor Fusion

Databases: MySQL, SQLite

Cloud/DevOps: AWS, Docker, Git, GitHub, CI/CD

Web/Tools: REST APIs, HTML, CSS, Tailwind CSS, VS Code, MATLAB

Platforms: GitHub Actions, Linux, Streamlit

EXPERIENCE

Full Stack Web Developer Intern – Wexor

One Month Internship

Full Stack Development

Coimbatore, Tamil Nadu

- Contributed to responsive web application development using modern frontend technologies, reusable components, and REST API integration.
- Worked across frontend and backend workflows, improving application functionality, responsive design, and maintainable full-stack implementation.

PROJECTS

Edge AI for Real-Time Predictive Monitoring of Power Transformers

[GitHub](#)

PyTorch — 1D CNN — GRU/TCN — Multimodal Sensor Fusion — Raspberry Pi — ONNX

- Designed a multimodal Edge AI system for transformer health monitoring using high-resolution electrical waveforms and thermal time-series data.
- Developed lightweight **1D CNN** and **GRU/TCN** branches with feature fusion for fault classification, anomaly scoring, and early fault-risk prediction.

AI Traffic Adaptation – Predictive Traffic Safety & Incident Intelligence

[GitHub](#)

PyTorch — Predictive Modeling — Computer Vision — Data Simulation — Model Evaluation

- Developed an AI-driven traffic intelligence system for **accident-risk prediction and near-miss identification** using traffic data.
- Built modular pipelines for data preparation, model development, traffic simulation, evaluation, testing, and predictive analysis.

Candida Auris – Infection Detection & Cause-of-Death Classifier

[GitHub](#)

XGBoost — SHAP — SMOTE — FastAPI — Healthcare Analytics

- Developed a multi-class ML pipeline for **cause-of-death classification** with preprocessing, scaling, missing-value handling, and SMOTE-based class balancing.
- Trained an **XGBoost** model with SHAP explainability and deployed predictions through FastAPI REST endpoints for single and batch inference.

Knowledge Assistant – RAG-Based AI Research Assistant

[GitHub](#)

LLMs — Retrieval-Augmented Generation — ChromaDB — Semantic Search — FastAPI

- Developed a RAG-based AI assistant for querying custom knowledge bases using document processing, embeddings, and semantic retrieval.
- Integrated **ChromaDB** for vector storage and similarity search with FastAPI services for application-level integration.

AREAS OF INTEREST

• AI/ML • Deep Learning • LLMs & RAG • Edge AI • Computer Vision • NLP • Data Engineering

CERTIFICATIONS

• MATLAB Certification Course • Flutter Bootcamp Certification • AI/ML & Generative AI Learning Programs

LEADERSHIP & ACTIVITIES

- Director** – Entrepreneurship Development Cell (EDC), CIT • **Joint Secretary** – EDC, CIT, 2025–2026
- Contributed to entrepreneurship initiatives, innovation challenges, ideathons, and technical events.

ACHIEVEMENTS

- Winner – Smart India Hackathon (SIH) Internal Hackathon, 2026** • **Winner – Bid to Build**
- 2nd Place – InnovatX 2026, Hackathon & Ideathon** • **Selected – TechnoMind 2K26 National-Level Project Exhibition**
- Top 50 Teams – National-Level Project Submission**